

JEE MAIN REVISION MASTER: VECTOR ALGEBRA

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 282 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Dot Product, Projections & Orthogonality	191 PYQs	Scalar Product
Very High	Cross Product & Geometric Areas	184 PYQs	Vector Product
Very High	Scalar Triple Product (Box Product) & Coplanarity	55 PYQs	Volume & Coplanarity
Very High	Section Formula, Collinearity & Centroids	38 PYQs	Geometry & Position Vectors
Very High	Vector Triple Product & Lagrange Identity	31 PYQs	Triple Products
Very High	Linear Combination & Vector Equations	30 PYQs	Vector Equations
High	Vector Algebra - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Dot Product, Projections & Orthogonality	$\vec{a} \cdot \vec{b} = \vec{a} \vec{b} \cos\theta$, $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$, $\text{Proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{ \vec{b} ^2} \vec{b}$
Cross Product & Geometric Areas	$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$, $\text{Area}_{\triangle} = \frac{1}{2} \vec{a} \times \vec{b} $, $\hat{n} = \frac{\vec{a} \times \vec{b}}{ \vec{a} \times \vec{b} }$
Scalar Triple Product (Box Product) & Coplanarity	$[\vec{a}; \vec{b}; \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$, $[\vec{a}; \vec{b}; \vec{c}] = 0 \iff \text{coplanar}$
Section Formula, Collinearity & Centroids	$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$, $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$, $\vec{a} = \lambda \vec{b} \iff \text{collinear}$
Vector Triple Product & Lagrange Identity	$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$, $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
Linear Combination & Vector Equations	$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$, $\text{or take } \vec{a} \times \vec{r} \text{ to decouple}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Dot Product, Projections & Orthogonality (Very High Yield — 191 PYQs)

Core Principle: Scalar product $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_x b_x + a_y b_y + a_z b_z$. Orthogonality condition: $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$. Scalar projection of $\vec{a}$ on $\vec{b}$: $\text{Proj}_{\vec{b}}(\vec{a}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$.

■ **Caution / JEE Pitfall:** 1. Vector projection of $\vec{a}$ along $\vec{b}$ is $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$. 2. $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2(\vec{a} \cdot \vec{b})$ (do not write $2|\vec{a}||\vec{b}|$ unless collinear!).

■ **Speed Trick:** Unit vector along the angle bisector of $\vec{a}$ and $\vec{b}$: $\vec{u} = \frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{a}=4\hat{i}-\hat{j}+3\hat{k}$, $\vec{b}=10\hat{i}+2\hat{j}-\hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b})+3\vec{c}=\vec{0}$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{u}$ and $\vec{v}$ be unit vectors inclined at an acute angle such that $|\vec{u} \times \vec{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \vec{u} + \vec{v}$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let P be a point in the plane of the vectors $\vec{AB}=3\hat{i}+\hat{j}-\hat{k}$ and $\vec{AC}=\hat{i}-\hat{j}+3\hat{k}$ such that P is equi..." [Ans: N/A]

3.2 Cross Product & Geometric Areas (Very High Yield — 184 PYQs)

Core Principle: Vector cross product $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta \hat{n}$. Collinearity condition: $\vec{a} \parallel \vec{b} \iff \vec{a} \times \vec{b} = \vec{0}$. Area of triangle with sides $\vec{a}$, $\vec{b}$ is $(1/2)|\vec{a} \times \vec{b}|$; parallelogram area is $|\vec{a} \times \vec{b}|$; parallelogram with diagonals $\vec{d}_1$, $\vec{d}_2$ is $(1/2)|\vec{d}_1 \times \vec{d}_2|$.

■ **Caution / JEE Pitfall:** 1. Cross product is anti-commutative: $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. 2. Lagrange's identity: $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$.

■ **Speed Trick:** Area of a parallelogram given diagonal vectors $\vec{d}_1$, $\vec{d}_2$ is $(1/2)|\vec{d}_1 \times \vec{d}_2|$, NOT $|\vec{d}_1 \times \vec{d}_2|$!

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{a}=4\hat{i}-\hat{j}+3\hat{k}$, $\vec{b}=10\hat{i}+2\hat{j}-\hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b})+3\vec{c}=\vec{0}$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{u}$ and $\vec{v}$ be unit vectors inclined at an acute angle such that $|\vec{u} \times \vec{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \vec{u} + \vec{v}$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let P be a point in the plane of the vectors $\vec{AB}=3\hat{i}+\hat{j}-\hat{k}$ and $\vec{AC}=\hat{i}-\hat{j}+3\hat{k}$ such that P is equi..." [Ans: N/A]

3.3 Scalar Triple Product (Box Product) & Coplanarity (Very High Yield — 55 PYQs)

Core Principle: Scalar triple product $[\vec{a} \ \vec{b} \ \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \det([\vec{a} \ \vec{b} \ \vec{c}])$. Three vectors are coplanar if and only if $[\vec{a} \ \vec{b} \ \vec{c}] = 0$. Volume of parallelepiped $V = |[\vec{a} \ \vec{b} \ \vec{c}]|$; tetrahedron volume $V_{\text{tet}} = (1/6)|[\vec{a} \ \vec{b} \ \vec{c}]|$.

■ **Caution / JEE Pitfall:** 1. Cyclic order preserves sign: $[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$. Non-cyclic swap negates: $[\vec{a} \ \vec{c} \ \vec{b}] = -[\vec{a} \ \vec{b} \ \vec{c}]$. 2. If any two vectors are identical or collinear, the box product is strictly 0.

■ **Speed Trick:** Four position points A, B, C, D are coplanar if and only if $[\vec{AB} \ \vec{AC} \ \vec{AD}] = 0$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} - 3\hat{k}$, $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{a}$..." [Ans: N/A]
- [2025 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with..." [Ans: N/A]
- [2025 Morning Shift] Q#3 (MCQ): "Let the position vectors of the vertices A, B and C of a tetrahedron ABCD be $\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + 3\hat{j} - 2\hat{k}$ and $2\hat{i} + \hat{j} + \hat{k}$..." [Ans: N/A]

3.4 Section Formula, Collinearity & Centroids (Very High Yield — 38 PYQs)

Core Principle: Section formula for point dividing line segment in ratio $m:n$: $\vec{r} = \frac{n\vec{b} + m\vec{a}}{m+n}$ (internal) and $\vec{r} = \frac{n\vec{b} - m\vec{a}}{n-m}$ (external). Centroid of triangle $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$. Three points are collinear if $\vec{AB} = \lambda \vec{BC}$.

■ **Caution / JEE Pitfall:** Collinear vectors can be antiparallel (negative scalar λ). External division formula carries an explicit minus sign in numerator and denominator: $\frac{n\vec{b} - m\vec{a}}{n-m}$.

■ **Speed Trick:** In a tetrahedron with vertices $\vec{a}, \vec{b}, \vec{c}, \vec{d}$, the centroid is $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c} + \vec{d}}{4}$.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "Let the lines $L_1: \vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$, $\lambda \in \mathbb{R}$ and $L_2: \vec{r} = (4\hat{i} + \hat{j} + \hat{k}) + \mu(2\hat{i} + 3\hat{j} + 4\hat{k})$, $\mu \in \mathbb{R}$..." [Ans: N/A]
- [2026 Evening Shift] Q#3 (MCQ): "Let O be the origin, $\vec{OP} = \vec{a}$ and $\vec{OQ} = \vec{b}$. If R is the point on $\vec{OP}$ such that $\vec{OP} = 5\vec{OR}$, and $\vec{OR} = \lambda \vec{a}$..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (Numerical): "Let PQR be a triangle such that $\vec{PQ} = -2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{PR} = a\hat{i} + b\hat{j} - 4\hat{k}$, $a, b \in \mathbb{Z}$. Let S be the p..." [Ans: 37]

3.5 Vector Triple Product & Lagrange Identity (Very High Yield — 31 PYQs)

Core Principle: Vector triple product expands via the BAC-CAB identity: $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$. It lies in the plane of $\vec{b}$ and $\vec{c}$, perpendicular to $\vec{a}$.

■ **Caution / JEE Pitfall:** Vector triple product is non-associative: $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$. Notice that $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$.

■ **Speed Trick:** Mnemonic: 'BAC minus CAB' — dot product of outer vectors multiplied by inner vector, minus dot product of first two multiplied by the third.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} - 3\hat{k}$, $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{a}$..." [Ans: N/A]
- [2025 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with..." [Ans: N/A]
- [2024 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = -5\hat{i} + \hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 4\hat{k}$ and $\vec{c} = ((\vec{a} \times \vec{b}) \times \vec{a}) \times \vec{b}$..." [Ans: N/A]

3.6 Linear Combination & Vector Equations (Very High Yield — 30 PYQs)

Core Principle: Solving vector equations like $\vec{r} \times \vec{a} = \vec{b}$ and $\vec{r} \cdot \vec{c} = d$. Express $\vec{r}$ as a linear basis expansion $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$ or take vector cross/dot products with known vectors on both sides.

■ **Caution / JEE Pitfall:** When $\vec{r} \times \vec{a} = \vec{b}$, note that $\vec{b} \perp \vec{a}$ (so $\vec{a} \cdot \vec{b} = 0$). The general solution is $\vec{r} = \lambda \vec{a} + \frac{\vec{a} \times \vec{b}}{|\vec{a}|^2}$.

■ **Speed Trick:** If $\vec{r} \times \vec{a} = \vec{b}$, taking the cross product with $\vec{a}$ gives $\vec{a} \times (\vec{r} \times \vec{a}) = \vec{a} \times \vec{b}$, decoupling $\vec{r}$ directly.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b}) + 3\vec{c} = \vec{0}$..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{u}$ and $\vec{v}$ be unit vectors inclined at an acute angle such that $|\vec{u} \times \vec{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \vec{u} \times \vec{v}$..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Let the vectors $\vec{a} = -\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$. For some $\lambda, \mu \in \mathbb{R}$, let $\vec{c} = \lambda\vec{a} + \mu\vec{b}$..." [Ans: N/A]

3.7 Vector Algebra - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Vector Algebra.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (Numerical): "If the distance of the point $P(43, \alpha, \beta)$, $\beta < 0$, from the line $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k})$, $\mu \in \mathbb{R}$ is..." [Ans: 170]
- [2025 Morning Shift] Q#3 (MCQ): "Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + 7\hat{j} + 3\hat{k}$. Let $\vec{r} = -\hat{i} + 2\hat{j} + \hat{k} + \lambda\vec{a} + \mu\vec{b}$..." [Ans: N/A]
- [2021 Evening Shift] Q#4 (Numerical): "For $p > 0$, a vector $\vec{v}_2 = 2\hat{i} + (p+1)\hat{j}$ is obtained by rotating the vector $\vec{v}_1 = \sqrt{3}p\hat{i} + \hat{j}$ by an angle..." [Ans: 6]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

If $\text{vec}(\mathbf{a}) = i\mathbf{j} + j\mathbf{k}$, $\text{vec}(\mathbf{b}) = j\mathbf{k}$ and $\text{vec}(\mathbf{c})$ be three vectors such that $\text{vec}(\mathbf{a}) \times \text{vec}(\mathbf{c}) = \text{vec}(\mathbf{b})$ and $\text{vec}(\mathbf{a}) \cdot \text{vec}(\mathbf{c}) = 3$, then $\text{vec}(\mathbf{c}) \cdot (\text{vec}(\mathbf{a}) - 2\text{vec}(\mathbf{b}))$ is equal to _____.

Correct Answer: 3

Solution Analysis: Let $\text{vec}(a) = i + j + k$, $\text{vec}(b) = j - k$. We have to find $\text{vec}(c) \cdot (\text{vec}(a) - 2\text{vec}(b))$ given that $\text{vec}(a) \times \text{vec}(c) = \text{vec}(b)$ and $\text{vec}(a) \cdot \text{vec}(c) = 3$. Let $\text{vec}(c) = xi + yj + zk$. Then $\text{vec}(a) = (1, 1, 1)$, $\text{vec}(b) = (0, 1, -1)$. Now compute $\text{vec}(a) \times \text{vec}(c)$: $\text{vec}(a) \times \text{vec}(c) = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ x & y & z \end{vmatrix}$. Expanding, $\text{vec}(a) \times \text{vec}(c) = i(z-y) - j(z-x) + k(y-x)$. Since this is equal to $\text{vec}(b) = j - k$, we compare components: $z-y=0$, $-(z-x)=1 \Rightarrow$

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Evening Shift (Q#2)

If the distance of the point $P(43, \alpha, \beta)$, $\beta < 0$, from the line $\text{vec}(r) = 4i - k + \mu(2i + 3k)$, $\mu \in \mathbb{R}$ along a line with direction ratios $3, -1, 0$ is $13\sqrt{10}$, then $\alpha^2 + \beta^2$ is equal to _____

Correct Answer: 170

Solution Analysis: Point P : $(43, \alpha, \beta)$ with the constraint $\beta < 0$. Target Line (L_1) : Given by $\text{vec}(r) = 4i - k + \mu(2i + 3k)$. A general point Q on this line is $(4 + 2\mu, 0, -1 + 3\mu)$. Direction of measurement : The distance is measured along a line with direction ratios $(3, -1, 0)$. Find Point Q on the Target Line Since the distance is measured along the direction $d = (3, -1, 0)$, the vector PQ must be parallel to d . $PQ = Q - P = (4 + 2\mu - 43, 0 - \alpha, -1 + 3\mu - \beta)$ $PQ = (2\mu - 39, -\alpha, 3\mu - \beta - 1)$ $PQ = (2\mu - 39, -\alpha, 3\mu - \beta - 1)$

Benchmark Problem 3: JEE Main 2026 (Online) 28th January Morning Shift (Q#3)

Let PQR be a triangle such that $\text{vec}(\text{PQ}) = -2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\text{vec}(\text{PR}) = a\mathbf{i} + b\mathbf{j} - 4\mathbf{k}$, $a, b \in \mathbb{Z}$. Let S be the point on QR, which is equidistant from the lines PQ and PR. If $|\text{vec}(\text{PR})| = 9$ and $\text{vec}(\text{PS}) = i - 7j + 2k$, then the value of $3a - 4b$ is _____

Correct Answer: 37

Solution Analysis: We are given that $|\text{vec}(\text{PR})|=9$ Using the distance formula for the vector $\text{PR}=\text{ai}+\text{bj}-4\text{k}$:
 $\sqrt{(\text{a}^2)+(\text{b}^2)+(-4)^2}=9 \Rightarrow \text{a}^2+\text{b}^2+16=81 \Rightarrow \text{a}^2+\text{b}^2=65$ The problem states that point S lies on QR and is equidistant from lines PQ and PR . In a triangle, the locus of points equidistant from two sides is the angle bisector. Therefore, $\text{vec}(\text{PS})$ is the internal angle bisector of $\angle \text{QPR}$. For an angle bisector $\text{vec}(\text{PS})$, it must be collinear with the sum of the unit vectors along the adjacent sides : $\text{vec}(\text{PS})=\lambda \frac{\text{vec}(\text{PQ})}{|\text{vec}(\text{PQ})|} + \lambda \frac{\text{vec}(\text{PR})}{|\text{vec}(\text{PR})|}$